AofGLP Book-Repo Harmonization Plan (v2)

Terminology

- **Module**: A `.glp` file
- **Program**: A top-level procedure (entry point) within a module that should be called to run/test

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Table 1: Modules in Both Book and Repo

Module	Book Location	Repo Path	Programs
Part I: Foundations			
`gates_simple.glp`	Ch4 constants	—	`and/3`, `or/3` `not/2`, `xor/` `nand/3`, `half_adder/4` `full_adder/5` `adder/4`
`gates.glp`	—	`constants/gates/gates.glp`	`and/3`, `or/3`
Part II: Streams			
`producer_consumer.glp`	Ch5 streams	`streams/producers_consumers/producer_consumer.glp`	`producer/1`, `consumer/2`
`reverse.glp`	Ch5 streams	`recursive/list_processing/reverse.glp`	`reverse/2`
`append.glp`	Ch5 streams	`recursive/list_processing/append.glp`	`append/3`
`fair_merge.glp`	Ch3 glp_core, Ch5	`streams/producers_consumers/fair_merge.glp`	`merge/3`
`monitor.glp`	Ch5 streams	`streams/objects_monitors/monitor.glp`	`monitor/1`

Module	Book Location	Repo Path	Programs
observed_monitor.glp	Ch5 streams	streams/objects_monitors/observed_monitor.glp	play_accum/
bounded_buffer.glp	Ch5 streams	streams/buffered_communication/bounded_buffer.glp	sq_num_buf
switch2x2.glp	Ch5 exercise	streams/buffered_communication/switch2x2.glp	switch2x2/4
Part II: Recursive			
plus.glp	Ch6 recursive	recursive/arithmetic_trees/plus.glp	plus/3, natural_num
times.glp	Ch6 recursive	recursive/arithmetic_trees/times.glp	times/3
factorial.glp	Ch6 recursive	recursive/arithmetic_trees/factorial.glp	factorial/2
fibonacci.glp	Ch6 recursive	recursive/arithmetic_trees/fibonacci.glp	fib/2
binary_tree.glp	Ch6 recursive	recursive/structure_processing/binary_tree.glp	binary_tree/
substitute.glp	Ch6 recursive	recursive/structure_processing/substitute.glp	substitute/4
observe.glp	Ch6 recursive	recursive/structure_processing/observe.glp	observe/3, test_observe
distribute.glp	Ch6 recursive	streams/producers_consumers/distribute.glp	distribute/3
distribute_nonground.glp	Ch6 recursive	recursive/structure_processing/distribute_nonground.glp	distribute/3, test_distribu
Part II: Meta			
plain_meta.glp	Ch7 meta	meta/plain/plain_meta.glp	run/1
failsafe_meta.glp	Ch7 meta	meta/plain/failsafe_meta.glp	run/2

Module	Book Location	Repo Path	Programs
control_meta.glp	Ch7 meta	meta/enhanced/control_meta.glp	run/2
tracing_meta.glp	Ch7 meta	meta/enhanced/tracing_meta.glp	run/3
Part III: Social Graph			
agent.glp	Ch8 social_graph	multiagent/social_graph/agent.glp	agent/3
cold_call.glp	Ch8 social_graph	multiagent/social_graph/cold_call_glpsam.glp	social_graph
response_handling.glp	Ch8 social_graph	multiagent/social_graph/response_handling_glpsam.glp	bind_respon handle_respo
friend_introduction.glp	Ch8 social_graph	multiagent/social_graph/friend_introduction_glpsam.glp	social_graph (intro clauses
network2.glp	Ch8 social_graph	multiagent/social_graph/network2.glp	network2/2
play_alice_bob.glp	Ch8 social_graph	multiagent/social_graph/play_alice_bob.glp	play_alice_b
play_introduction.glp	Ch8 social_graph	multiagent/social_graph/play_introduction.glp	play_introdu
Part III: Social Networks			
play_dm.glp	Ch9 social_networks	multiagent/social_networks/play_dm.glp	play_dm/0
play_feed.glp	Ch9 social_networks	multiagent/social_networks/play_feed.glp	play_feed/0
play_child_safe.glp	Ch9 social_networks	multiagent/social_networks/play_child_safe.glp	play_child_s
interlaced_streams.glp	Ch10 interlaced	multiagent/social_networks/interlaced_streams.glp	streams/2

Module	Book Location	Repo Path	Programs
Library (Appendix)			
`channel_ops.glp`	Library appendix	`lib/channels/channel_ops.glp`	`send/3`, `rec` `new_channe`
`relay.glp`	Library appendix	`lib/channels/relay.glp`	`relay/3`
`lookup.glp`	Library appendix	`lib/lookup/lookup.glp`	`lookup/3`, `lookup_send`
`tag_stream.glp`	Library appendix	`lib/streams/tag_stream.glp`	`tag_stream/3`
`inject.glp`	Library appendix	`lib/streams/inject.glp`	`inject/4`
`broadcast.glp`	Library appendix	`lib/broadcast/broadcast.glp`	`broadcast/3`
`time_utils.glp`	Library appendix	`lib/time/time_utils.glp`	`current_time`

Table 2: Modules in Book NOT in Repo

Module	Book Location	Programs	Action
`gates_simple.glp`	Ch4 constants	`and/3`, `or/3`, `not/2`, `xor/3` (facts)	CREATE
`circuits.glp`	Ch4 constants	`nand/3`, `half_adder/4`, `full_adder/5`, `adder/4`	CREATE
`lesseq.glp`	Ch6 recursive	`lesseq/2`	CREATE (or add to `min.glp`)
`flatten.glp`	Ch6 recursive	`flatten/2`	CREATE
`tree_sum.glp`	Ch6 recursive	`tree_sum/2`	CREATE
`list_to_bst.glp`	Ch6 recursive	`list_to_bst/2`	CREATE
`replay.glp`	Ch7 meta	`replay/3`	CREATE (or add to tracing_meta.glp)

Module	Book Location	Programs	Action
Exercise Solutions			
biased_merge.glp	Solutions appendix	bmerge/6	CREATE
length.glp	Solutions appendix	length/2	EXISTS but helper naming differs. FIX
map_inc.glp	Solutions appendix	map_inc/2	CREATE
filter_even.glp	Solutions appendix	filter_even/2	CREATE

Table 3: Modules in Repo NOT in Book

To REMOVE or MOVE from AofGLP/

Module	Repo Path	Reason	Action
cp_meta.glp	concurrent_prolog_examples/	CP syntax, not GLP	MOVE outside AofGLP
debugger_meta.glp	concurrent_prolog_examples/	CP syntax	MOVE outside AofGLP
*.pl (27 files)	from_aop/	Prolog, not GLP	MOVE outside AofGLP

To RENAME (drop _glpsam suffix)

Current Name	New Name	Repo Path
agent_glpsam.glp	agent_simple.glp	multiagent/social_graph/
cold_call_glpsam.glp	cold_call.glp	multiagent/social_graph/
response_handling_glpsam.glp	response_handling.glp	multiagent/social_graph/
friend_introduction_glpsam.glp	friend_introduction.glp	multiagent/social_graph/

Candidates for Book Inclusion

Module	Repo Path	Programs	Recommendation
`counter.glp`	`streams/objects_monitors/`	`counter/2`	Good for exercises (simple monitor)
`many_counters.glp`	`streams/objects_monitors/`	`use_many_counters/2`	Exercise solution candidate
`network_switch.glp`	`streams/objects_monitors/`	`network/2`	May be `network2` in book
`network_switch_3way.glp`	`streams/objects_monitors/`	`network/3`	Exercise solution
`mwm.glp`	`streams/producers_consumers/`	`mwm/2`	Complex - DISCUSS
`distribute_binary.glp`	`streams/producers_consumers/`	`distribute_binary/3`	Exercise in book
`distribute_indexed.glp`	`streams/producers_consumers/`	`distribute_indexed/3`	DISCUSS
`parallel_table.glp`	`streams/producers_consumers/`	`table/4`	Exercise in book
`merge_tree.glp`	`streams/producers_consumers/`	`merge_tree/2`	DISCUSS
`abortable_meta.glp`	`meta/enhanced/`	`run/3`	Could extend Ch7
`termination_detection_meta.glp`	`meta/enhanced/`	`run/3`	Could extend Ch7
`primes.glp`	`recursive/arithmetic_trees/`	`primes/2`	Classic example
`hanoi.glp`	`recursive/arithmetic_trees/`	`hanoi/5`	Classic puzzle
`quicksort.glp`	`recursive/list_processing/`	`quicksort/2`	In book text
`insertion_sort.glp`	`recursive/list_processing/`	`insertion_sort/2`	Good comparison
`merge_sort.glp`	`recursive/list_processing/`	`merge_sort/2`	In book exercises
`bubble_sort.glp`	`recursive/list_processing/`	`bubble_sort/2`	In book exercises

Variants to Consolidate

Variants	Keep	Archive
`replicate.glp`, `replicate2.glp`, `replicate3.glp`	DISCUSS which is best	Archive others

Variants	Keep	Archive
`play_4agent.glp`, `play_4agents.glp`	MERGE or clarify difference	
`observe.glp`, `observe_minimal.glp`, `observe_play.glp`	Keep all (different purposes)	

Supporting Modules (Keep)

Module	Repo Path	Purpose
`network3.glp`, `network4.glp`	`multiagent/social_graph/`	N-agent network switches for plays
`monitor_test.glp`	`streams/objects_monitors/`	Test harness for monitor
`play_*.glp` files	various	Test scenarios
`*_test.glp` files	various	Test harnesses

Summary Statistics

Current State

- **Modules in both**: ~35
- **Modules in book only**: ~12
- **Modules in repo only**: ~50 (including variants, tests, non-GLP)

After Harmonization

- **Shared modules**: ~45
- **Modules to create**: 12
- **Modules to remove/move**: ~30 (Prolog files, CP examples)
- **Modules to rename**: 4

Action Items by Priority

P1: Critical Fixes (Repo)

1. **Meta-interpreters**: Add `M#` module calls to all meta-interpreters

2. **Rename**: Drop `_glpsam` suffix from 4 files
3. **Move out**: `from_aop/` and `concurrent_prolog_examples/` outside AofGLP
4. **Fix helpers**: `reverse/3` → `reverse_acc/3`, `length/3` → `length_acc/3`

P2: Critical Fixes (Book)

1. **Add guards**: `substitute/4` needs `ground` guards
2. **Harmonize factorial**: Align base cases and guards

P3: Missing Modules (Create in Repo)

1. `gates_simple.glp` - simple fact-based gates
2. `circuits.glp` - nand, half_adder, full_adder, adder
3. `flatten.glp`, `tree_sum.glp`, `list_to_bst.glp`
4. Exercise solutions: `biased_merge.glp`, `map_inc.glp`, `filter_even.glp`

P4: Both Versions Needed

1. Producer/consumer: countdown (book) + count-up (repo) in both
2. Gates: simple facts (Ch4) + streams (later) in both

P5: Discuss

1. `observe/3` divergence - which version is canonical?
2. Which repo-only programs to add to book?
3. `replicate*.glp` variants - which to keep?